

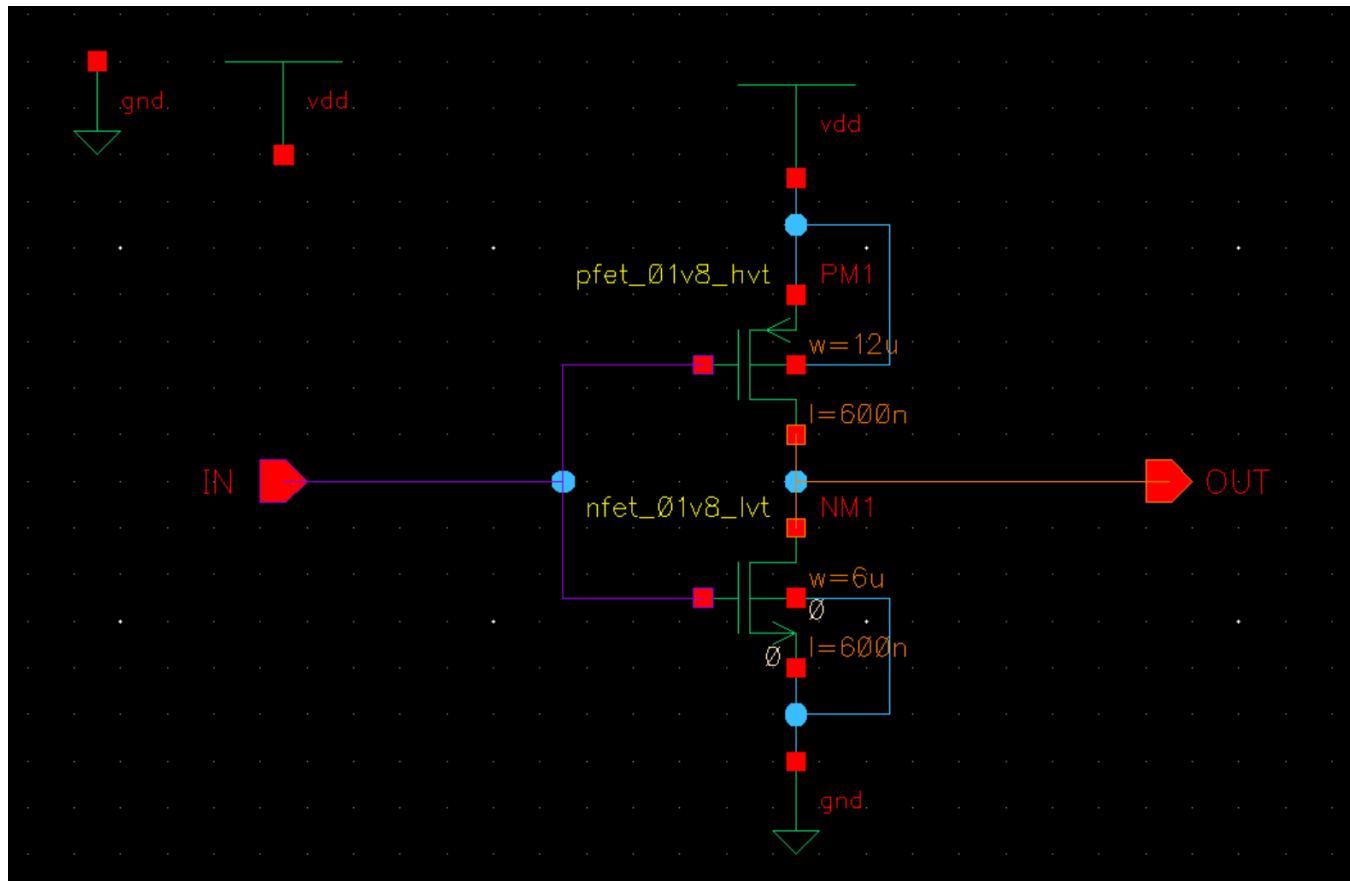
First Design Project

Colin Gregg

Overview

- Latch Inverter
- Transmission (pass) Gate
- Latch D-1 & Latch D-2
- Flip-Flop-D
- Shift Register – 4-bits
- CoutBAR
- SumBAR
- Adder Cell
- Shift Register - 1-bit (Static Latch)
- Full Serial Adder

Latch Inverter



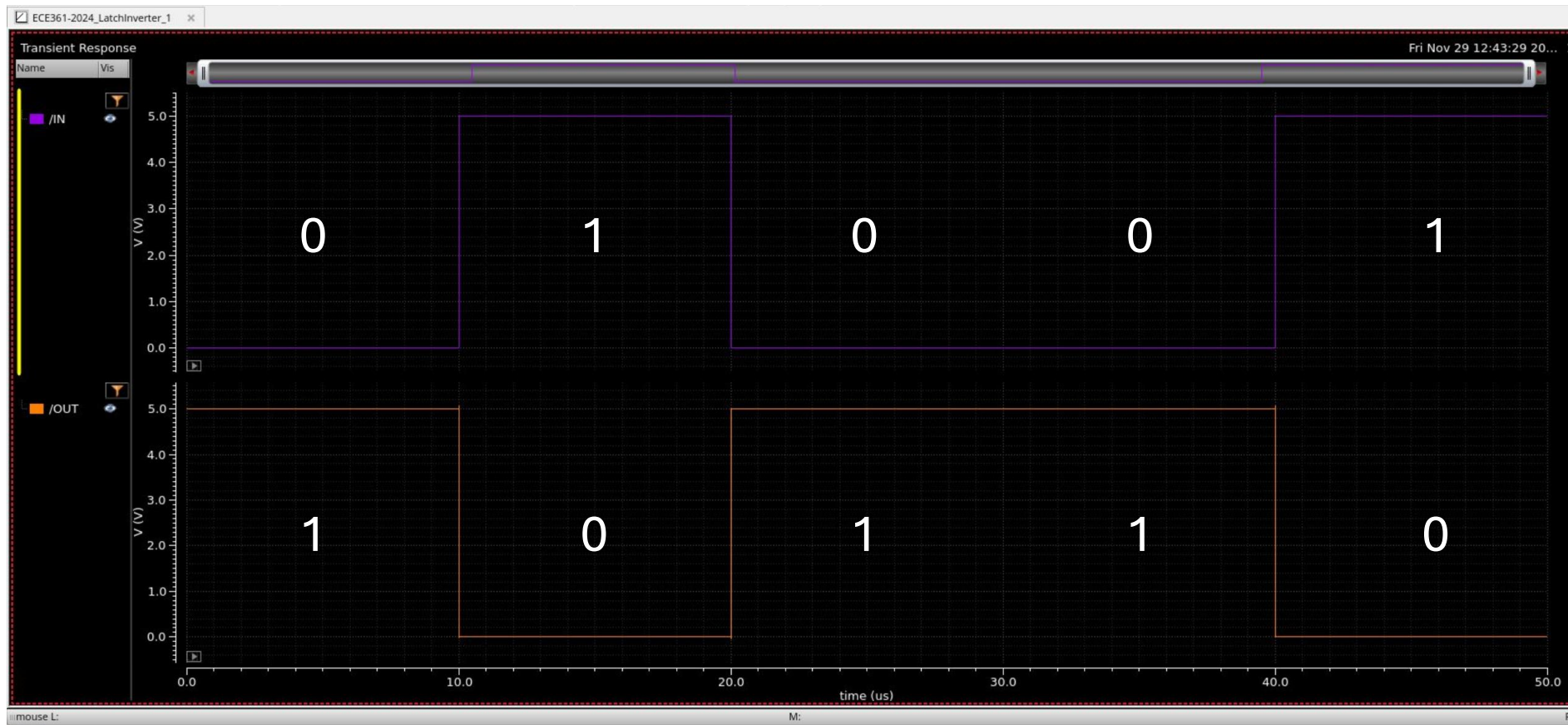
- Basic width for NMOS and PMOS for whole serial adder is 6microns and 12 microns, respectively



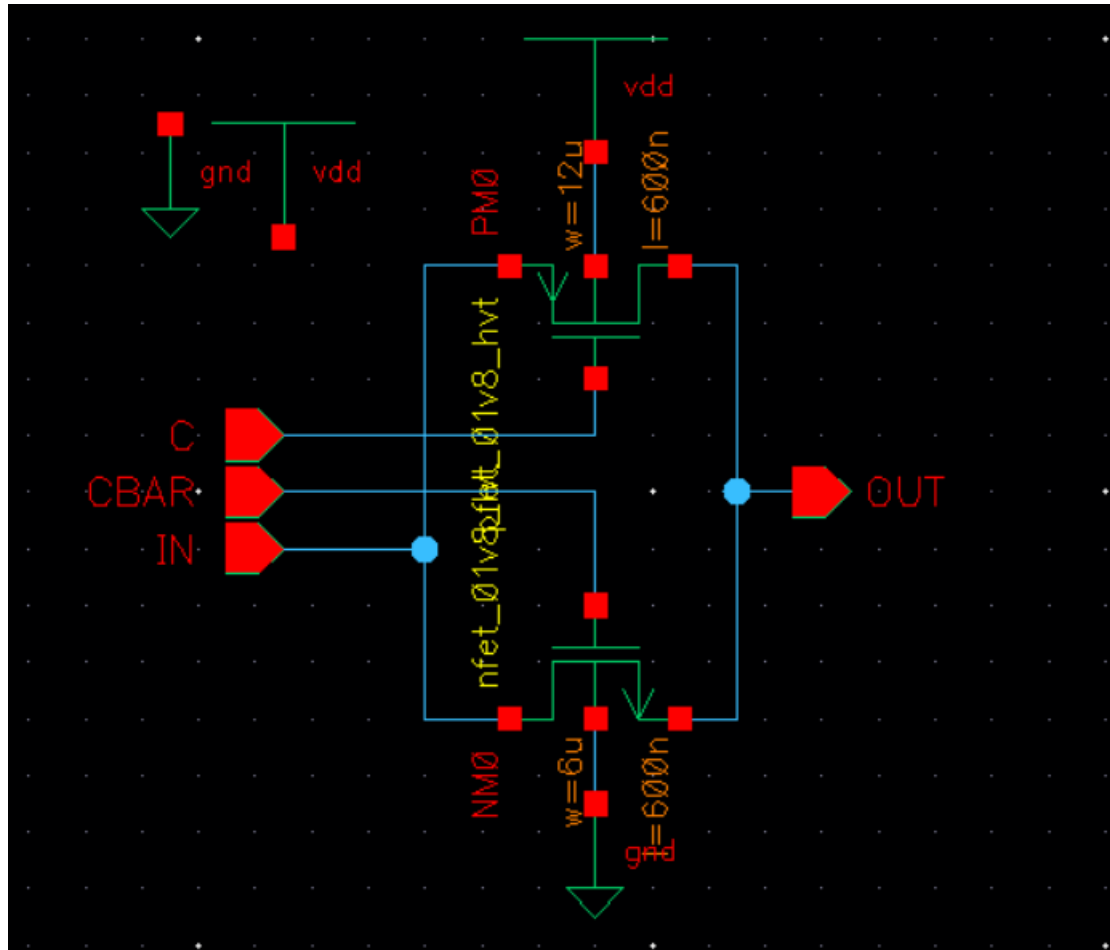
Latch Inverter - Simulation

Input:

Output:

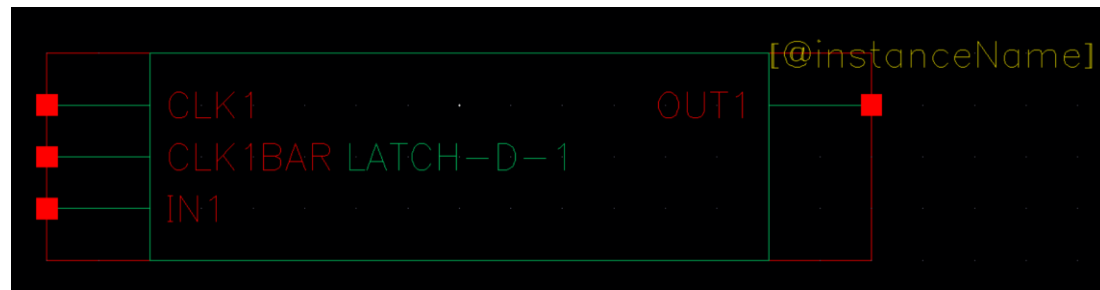
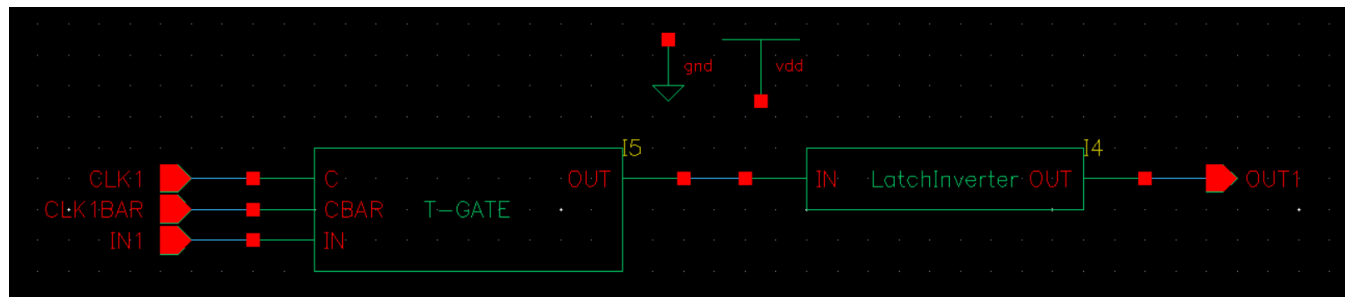


Transmission (pass) Gate

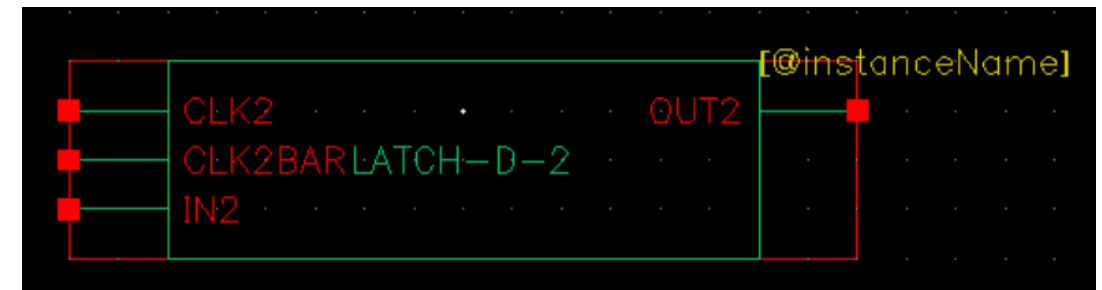
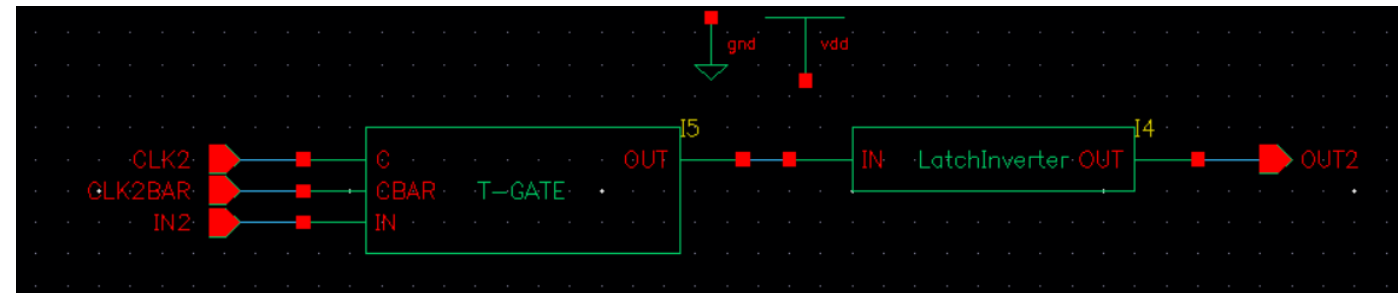


- NMOS drain connected to PMOS source
- Passes strongly on **LOW**

Latch D-1

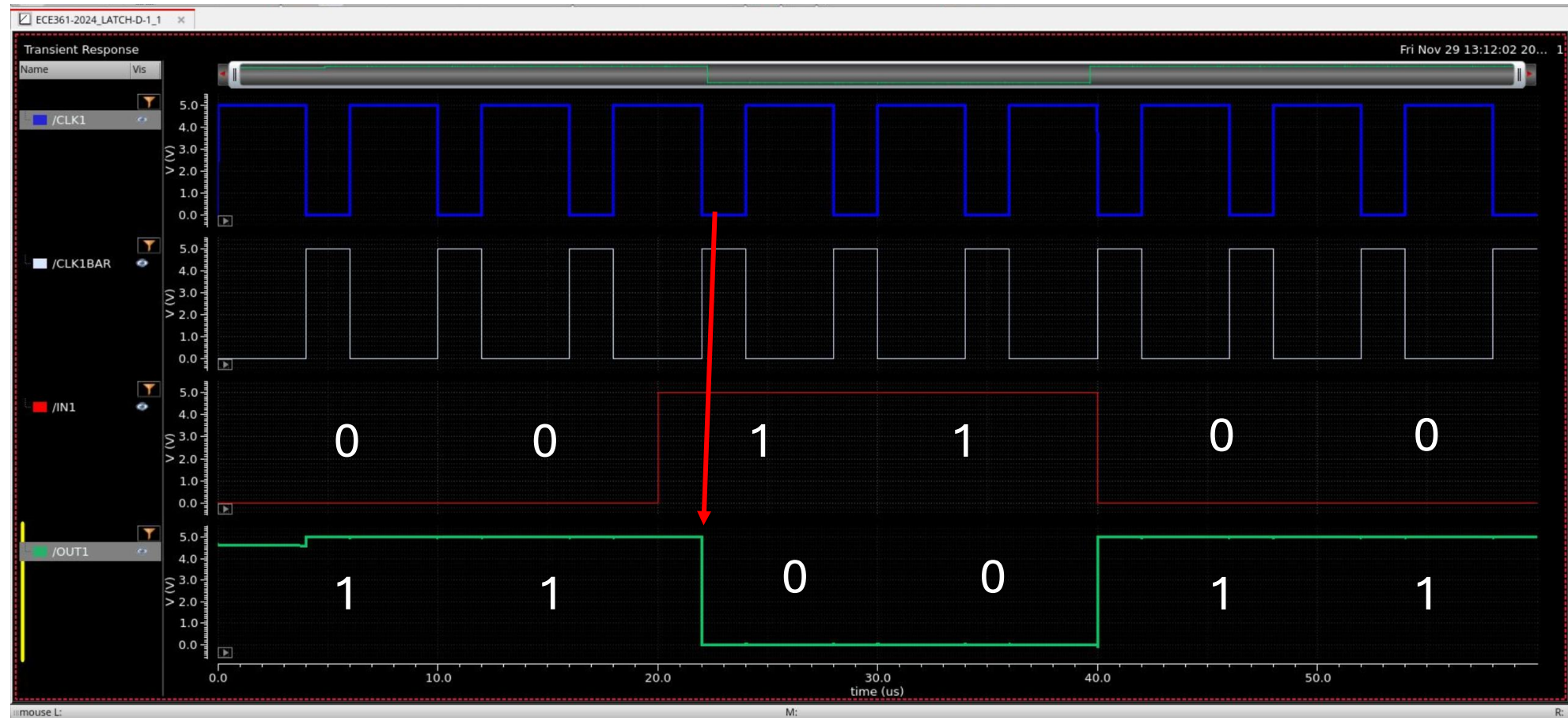


Latch D-2



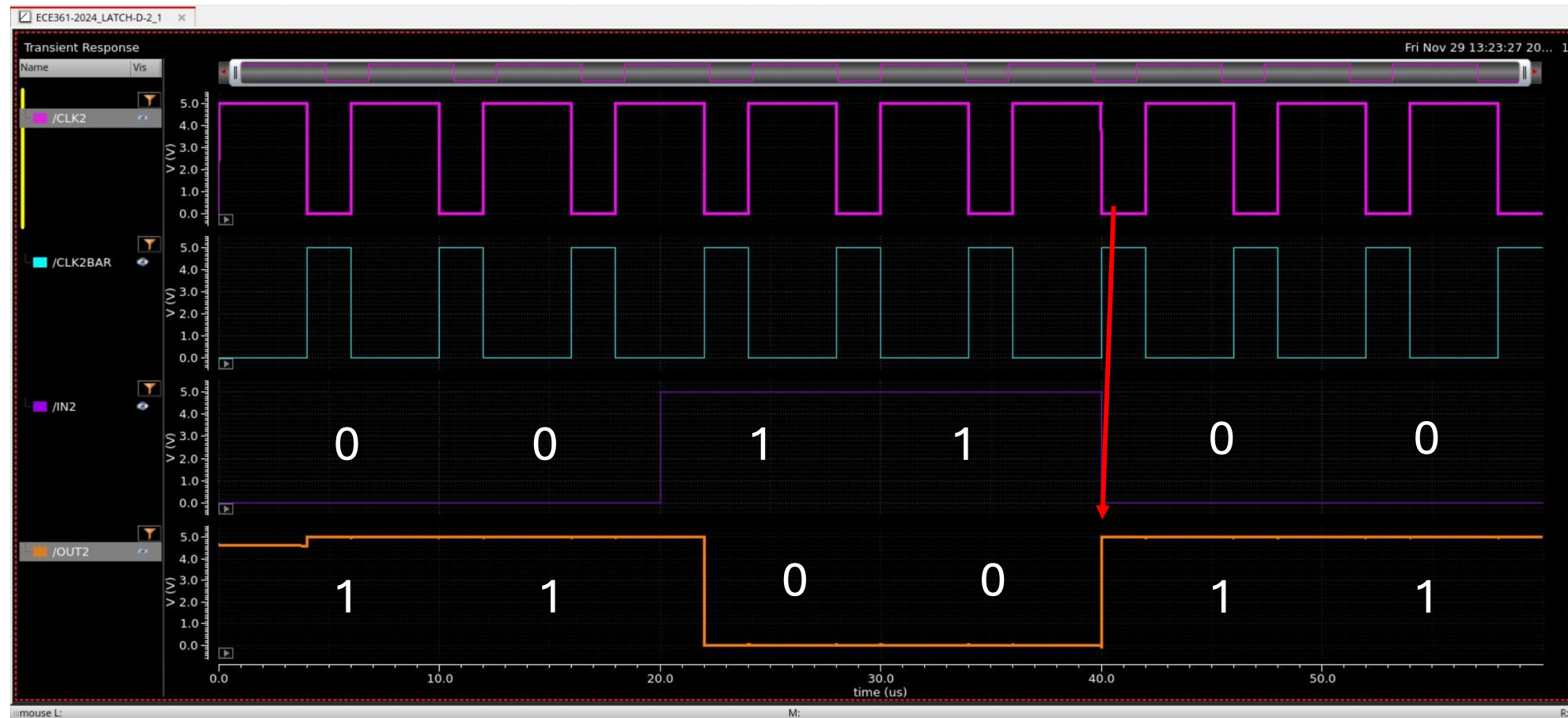
Latch D-1

We can see that as the input goes high and the clock is also high, the input is not able to pass through. However, once the clock goes low the T-gate allows the input to be passed through to the inverter, outputting a low value.



Latch D-2

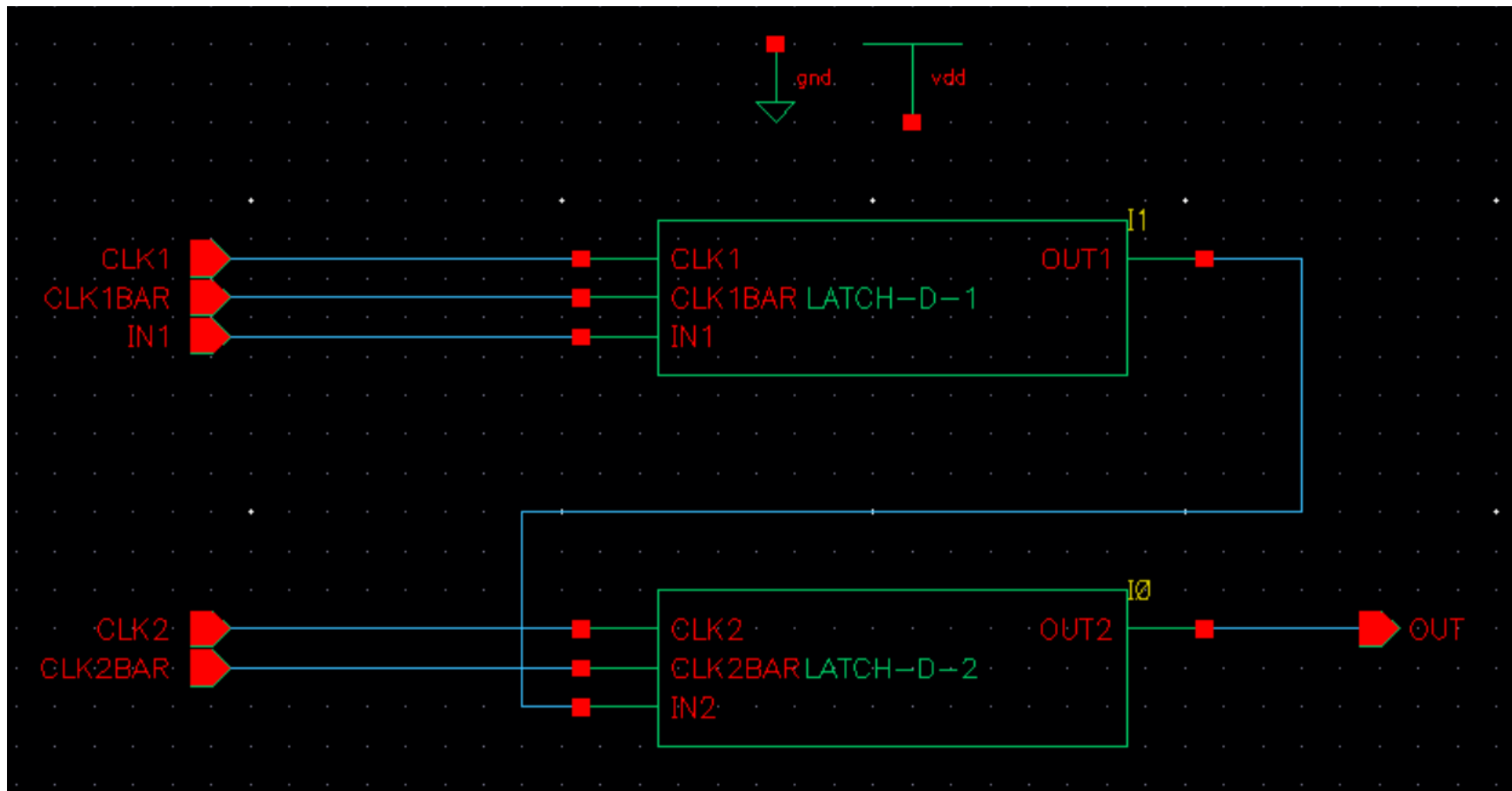
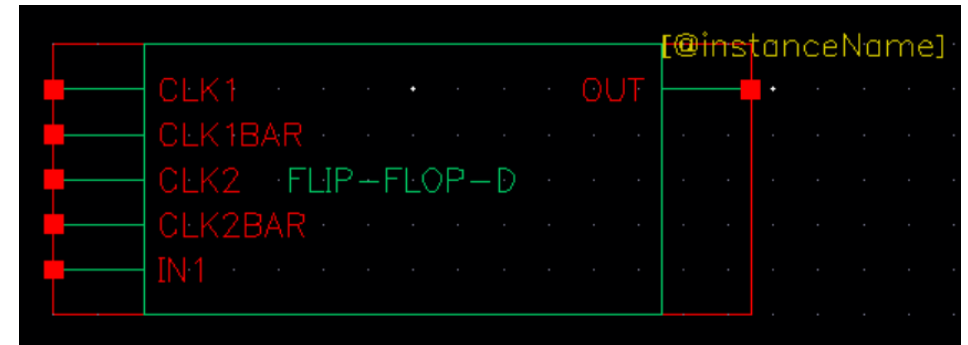
The same thing from Latch D-1 applies here. We can also see that when the input changes back to low, it does so while the clock is low so the input immediately passes through.



Latches...

- The latches work to store data because there is charge built up on the gates of the transistors in the inverter.
- These latches are dynamic in that the charge will decay due to reverse bias currents in the mosfets, and the stored value needs to be updated before that happens.
- We can also note that latches are transparent when the clock is low since the output directly sees the input.

Flip-Flop-D



- When CLK1 is low, the input will pass through latch 1.
- Then, when CLK2 is low it will pass through the second latch.

Flip-Flop-D: Simulation

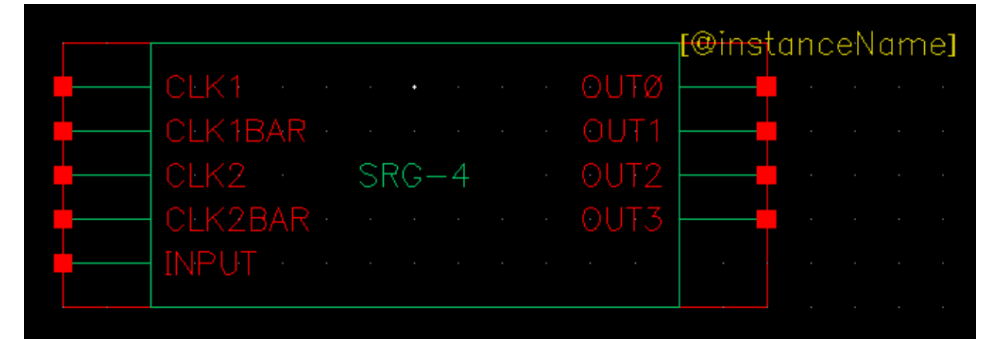
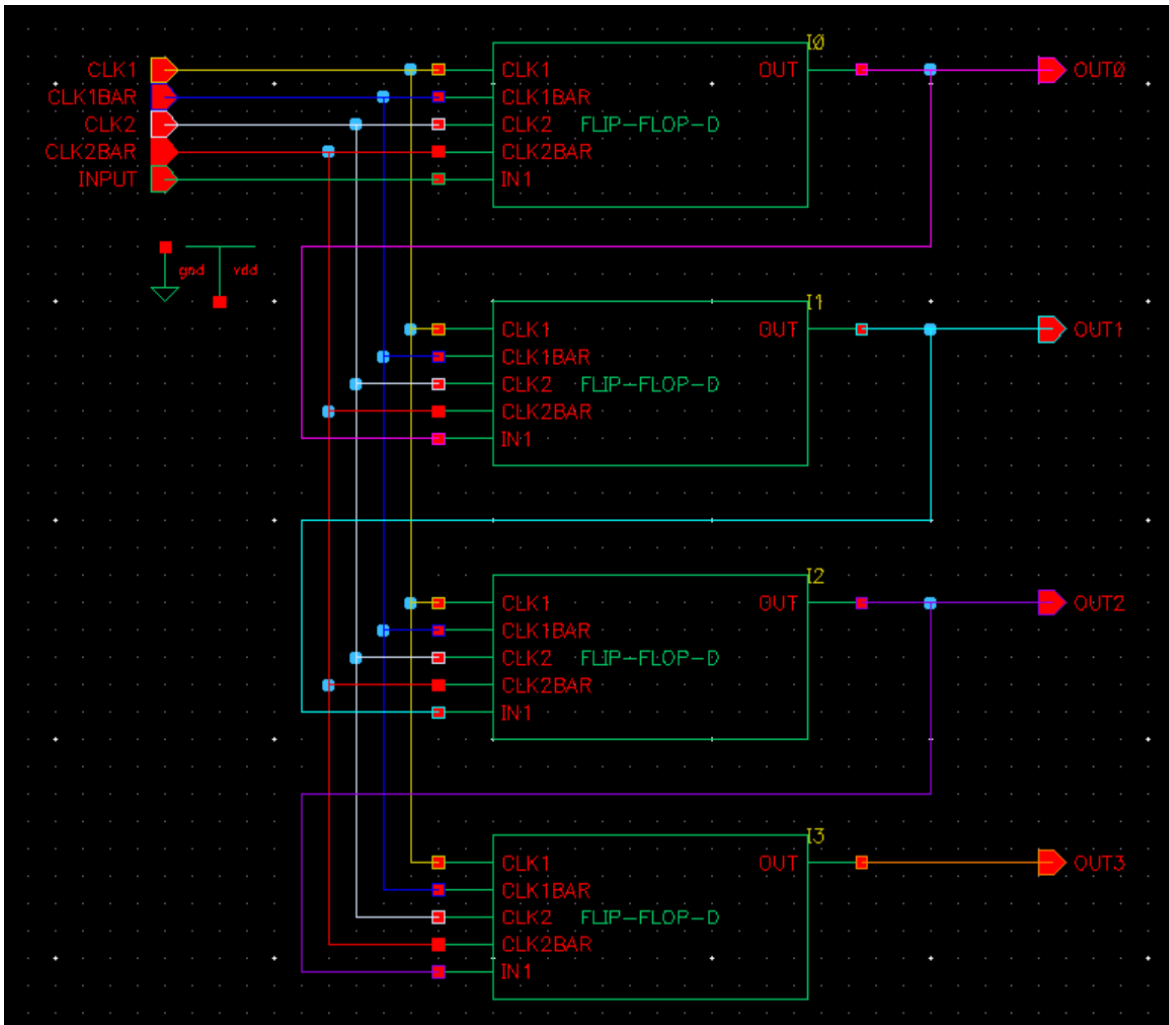
Clocks 1 and 2 need to both go low after the input has changed for the output to be updated



Dynamic Flip-Flop...

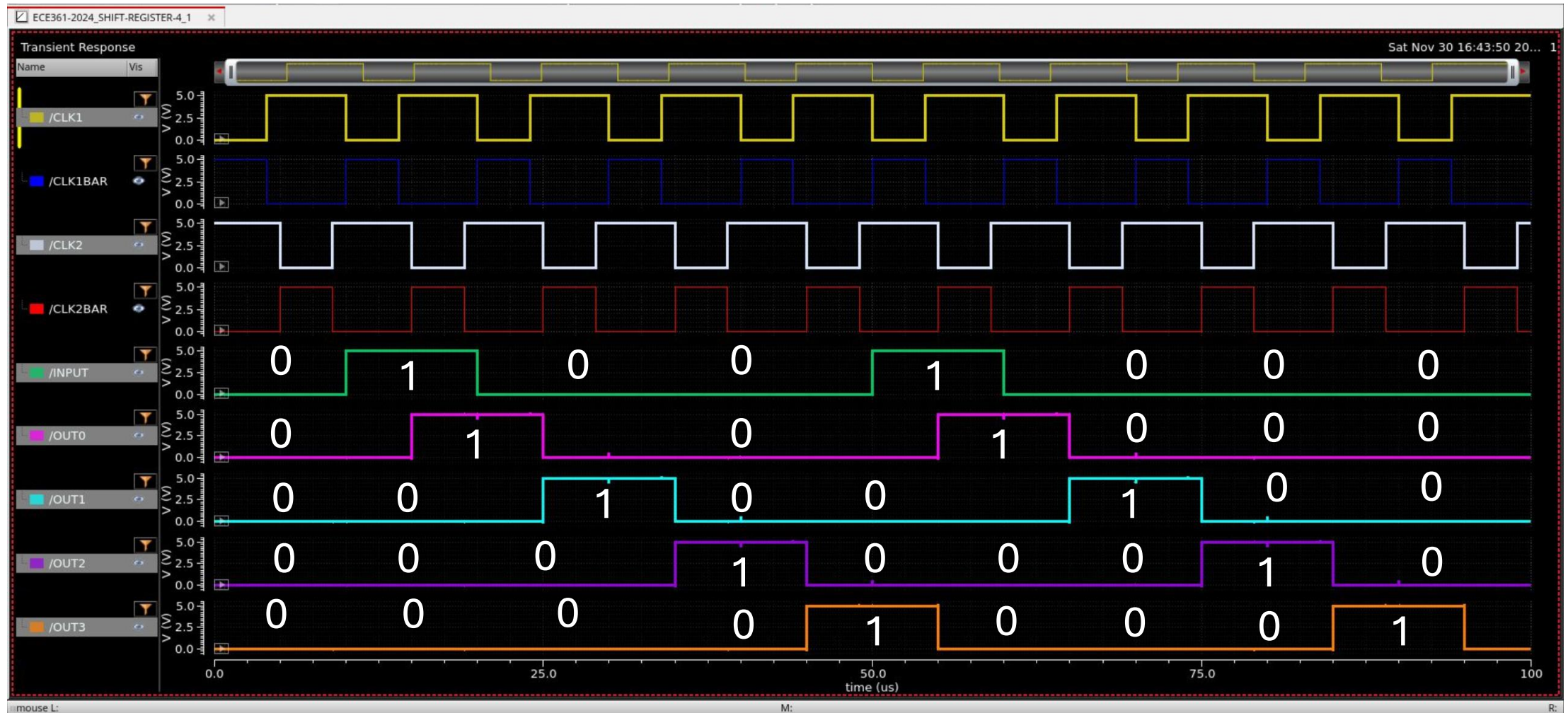
- This is a dynamic flip-flop because it is created from two dynamic latches
- Also, because it is created from two latches, two non-overlapping clocks are required, and the input does not pass to the output until both clocks have gone low.
- This means the flip-flop is not transparent.

4 Bit Shift Register



Create from 4 concurrent Flip-Flops which each delay the output by one clock cycle.

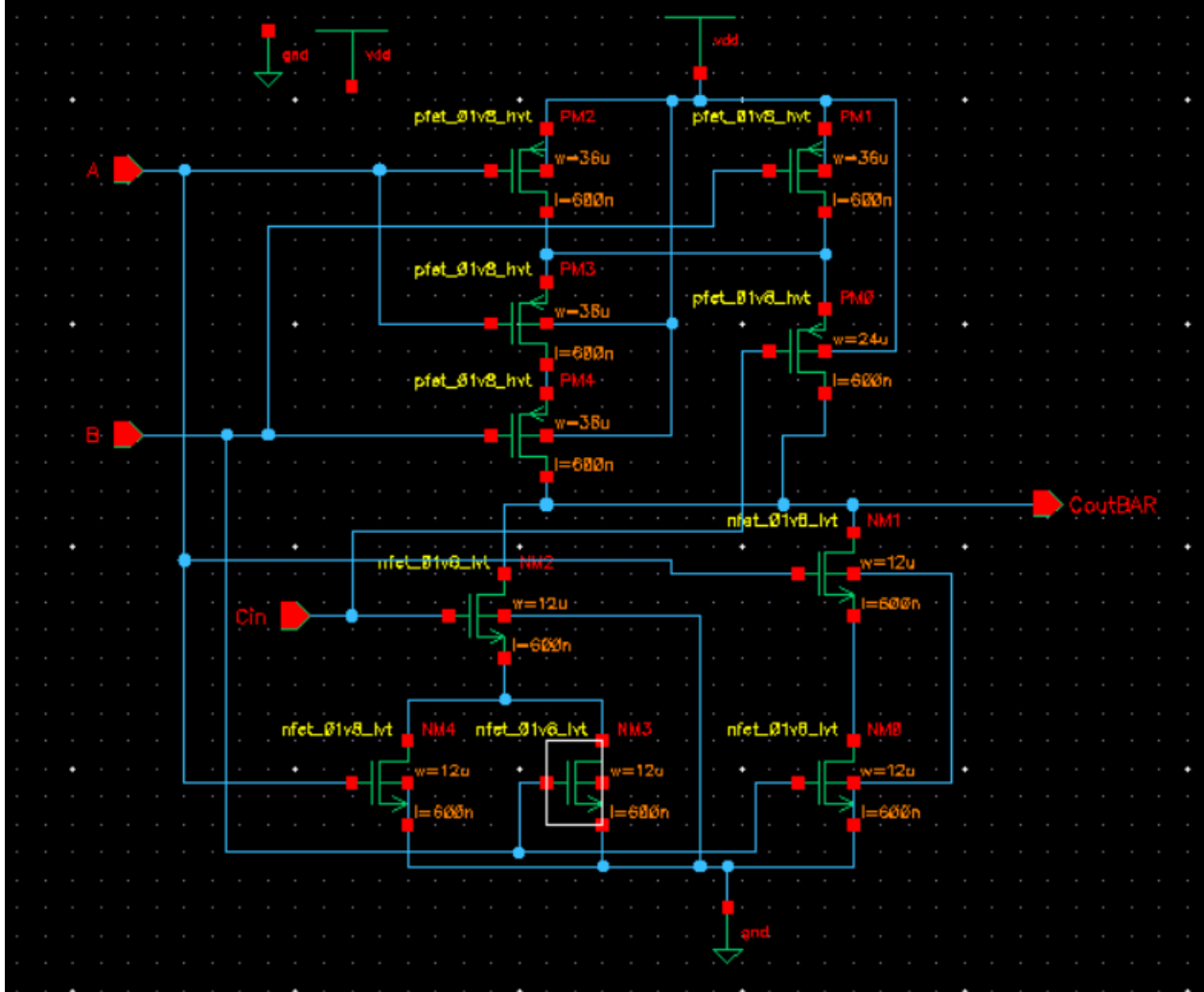
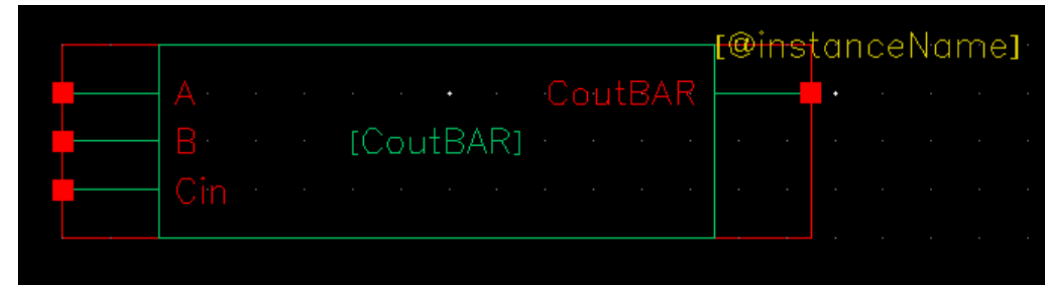
4 Bit Shift Register: Simulation



4 Bit Shift Register...

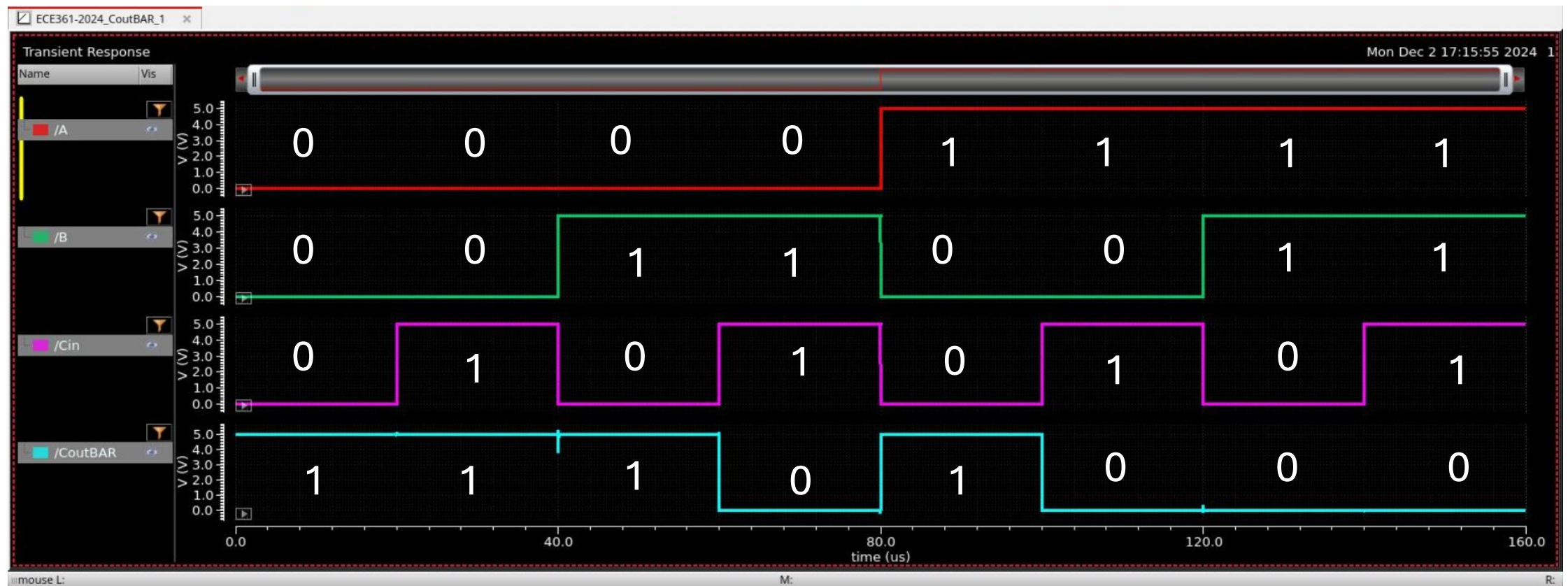
- We can see that the first output comes half a clock cycle after the input goes high.
- That is due to how the clocks are timed.
- Afterwards, the bit shifts rightward to the next output with the passing of each clock cycle.

Carry-out Bit



Because there are transistors in series, must resize them to properly adhere to rise and fall times of the NMOS and PMOS transistors.

Carryout Bit: Simulation



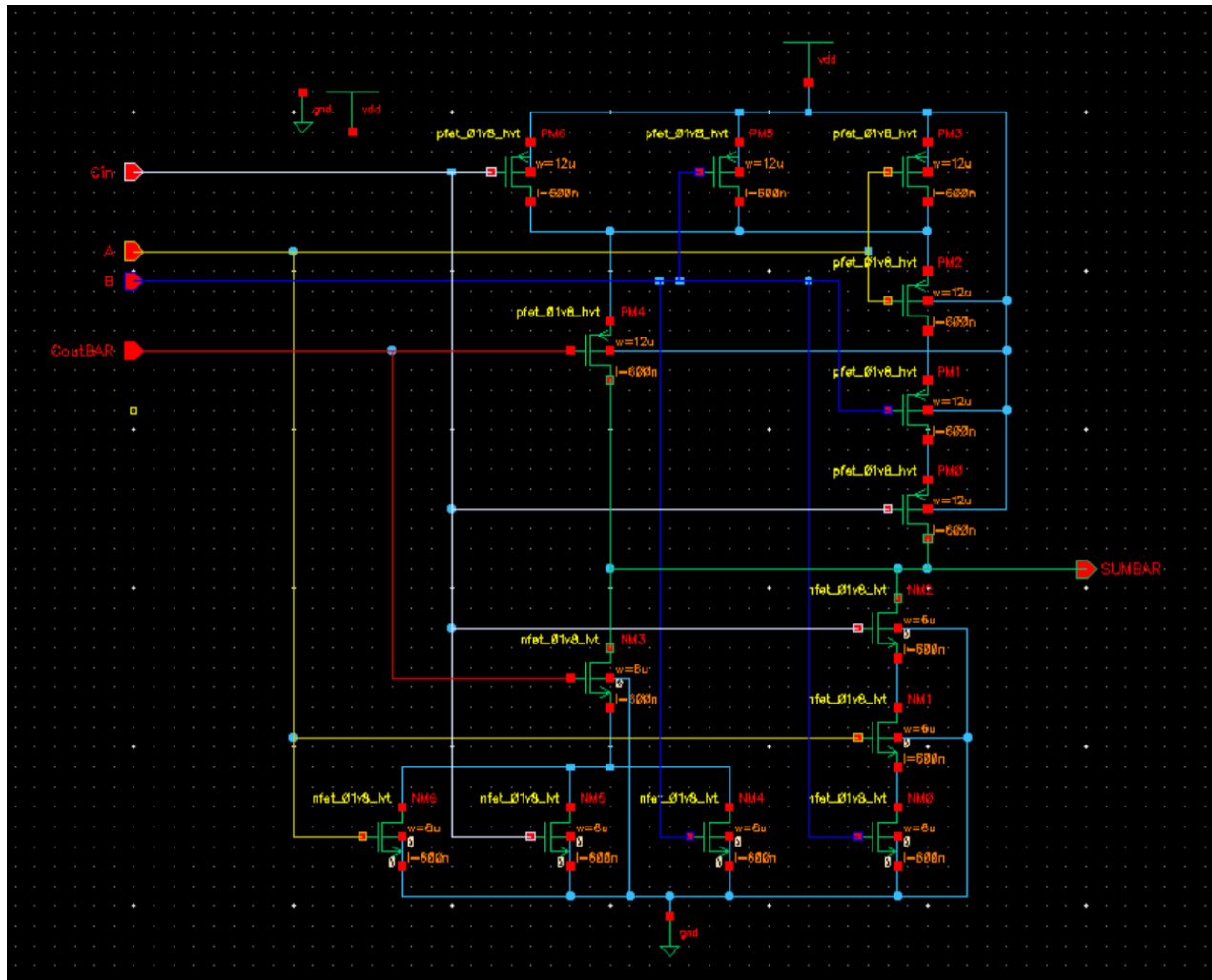
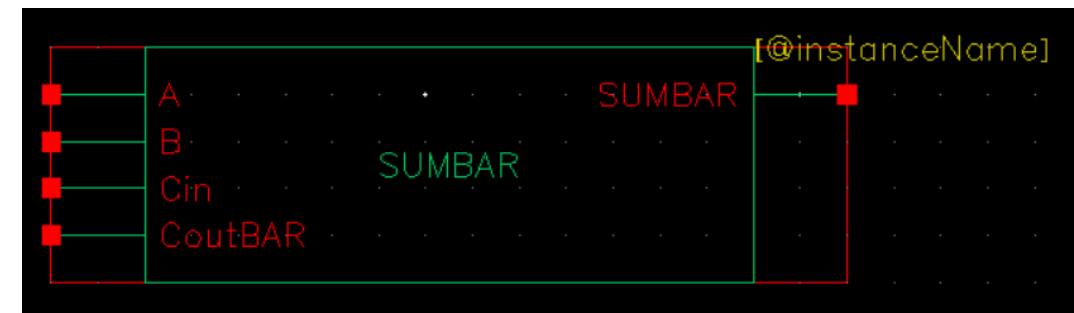
Carryout Bit: Truth Table

$$\text{Cout} = A*B + (A+B)*\text{Cin}$$

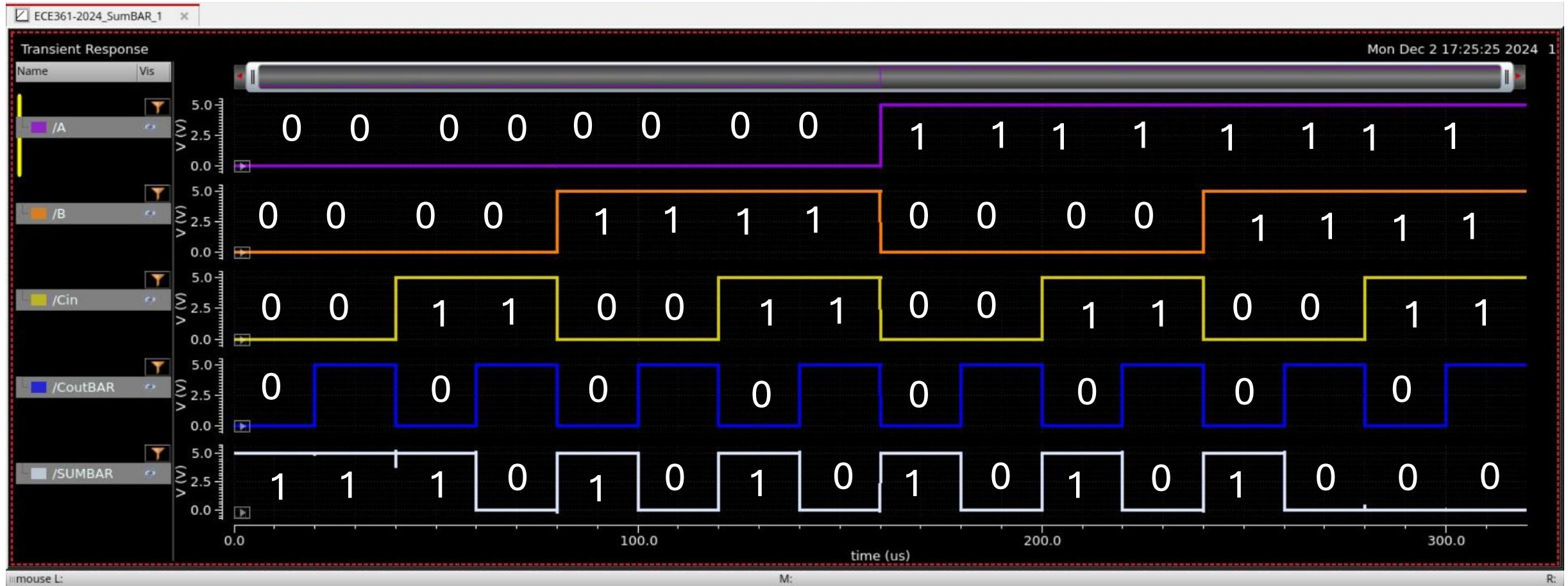
A	B	Cin	Cout
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

The simulation results exactly match the opposite of the Cout value taken from the its equation.

SumBAR



SumBAR: Simulation

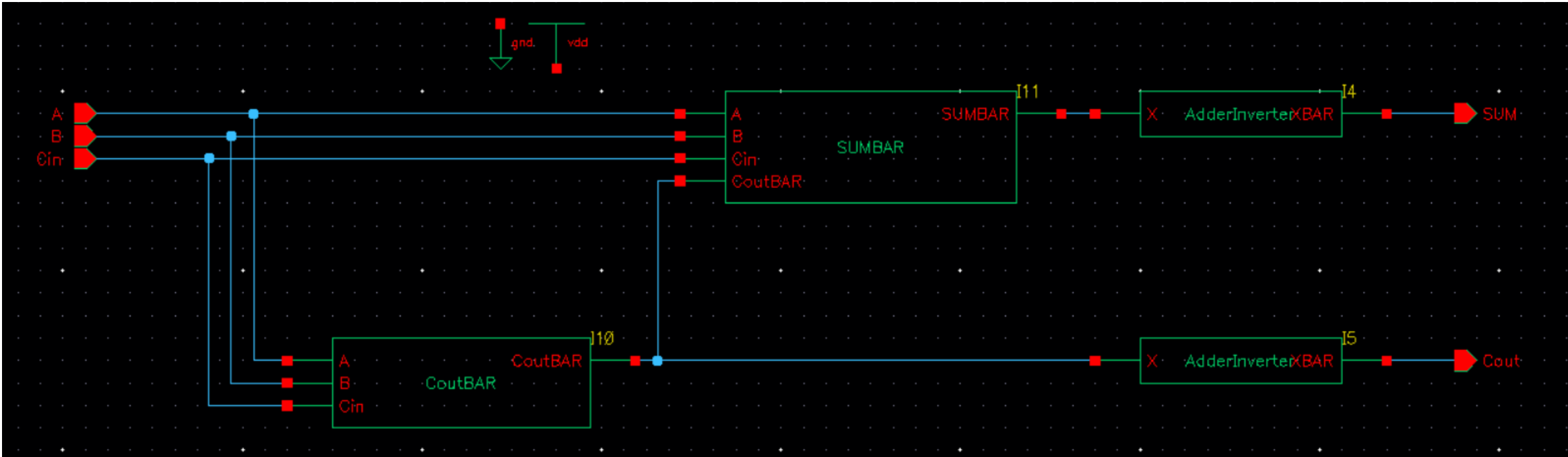
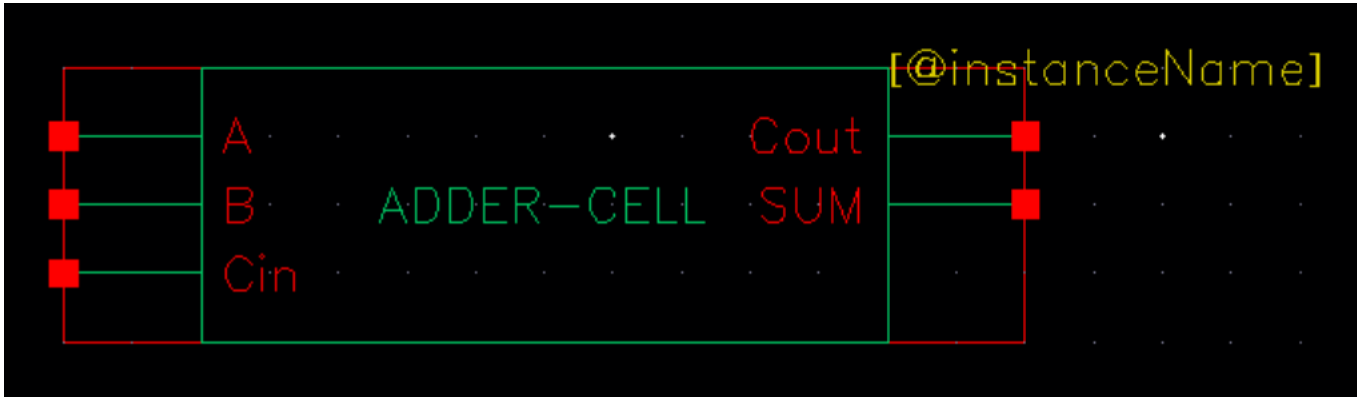


SumBAR: Truth Table...

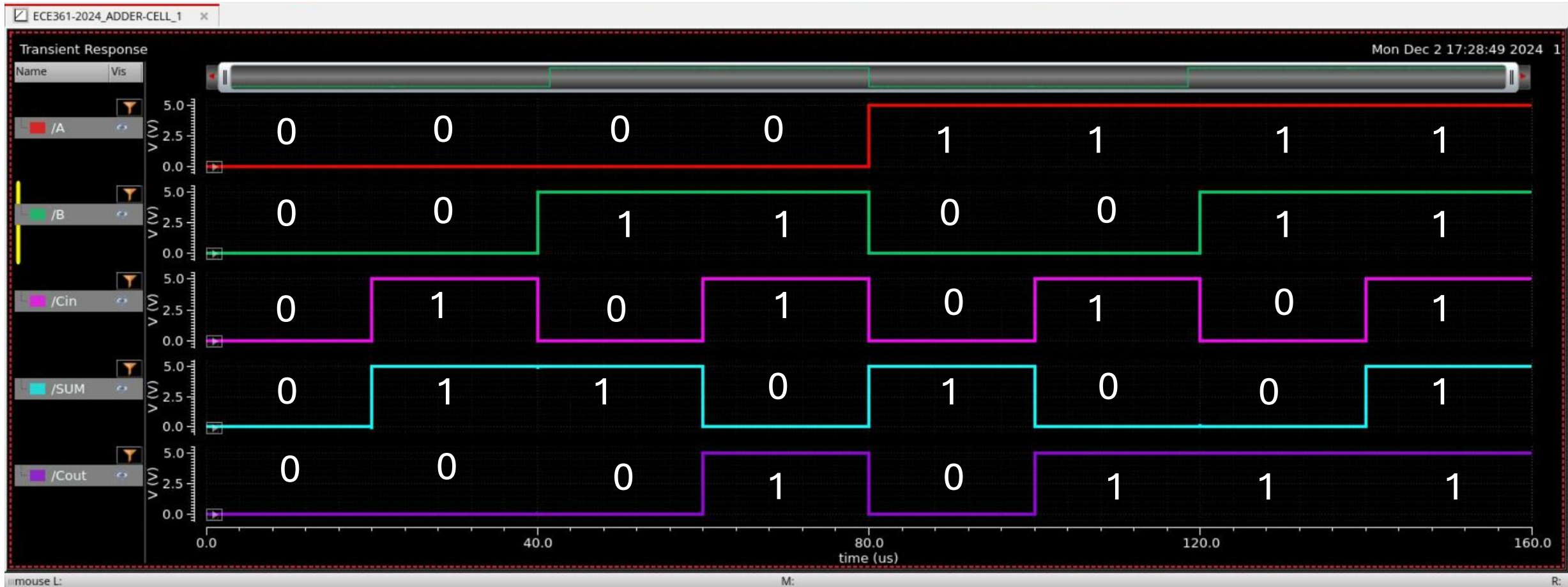
A	B	Cin	~Cout	SUM
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

Matches exactly
opposite with the
simulation results.

Adder Cell



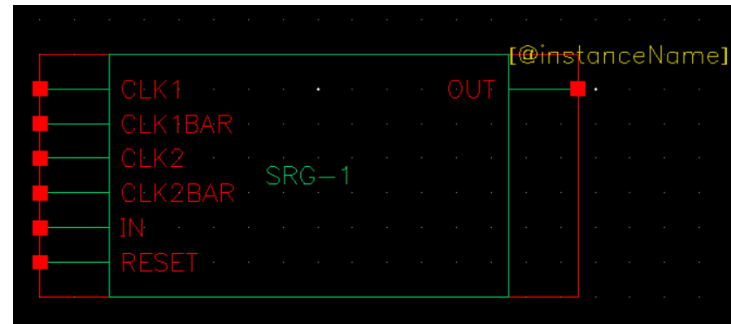
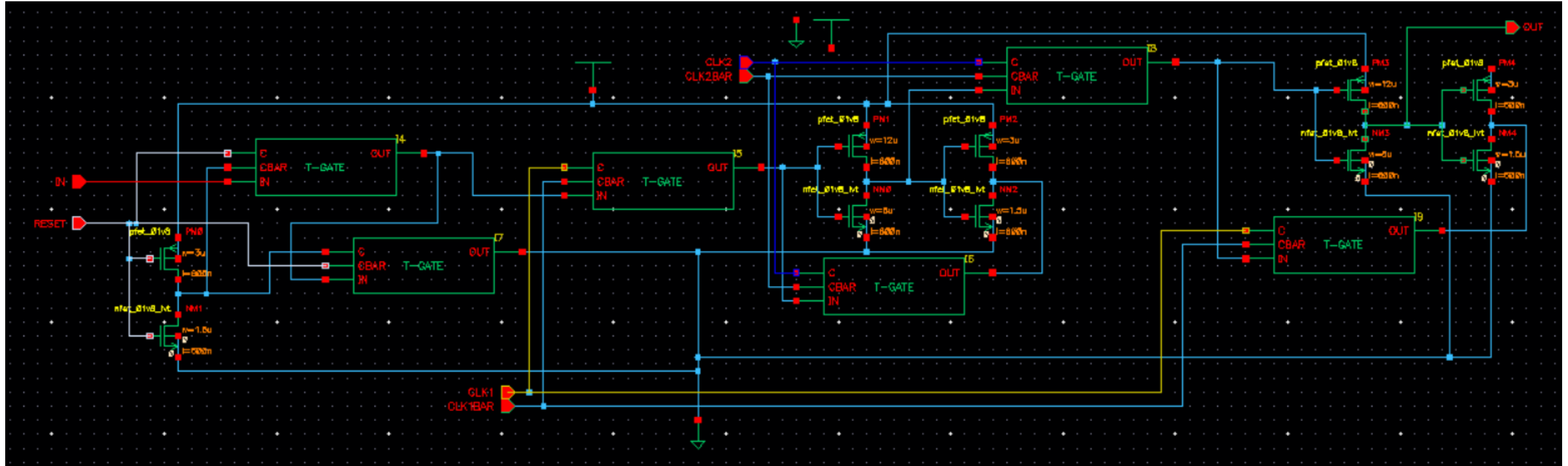
Adder Cell: Simulation



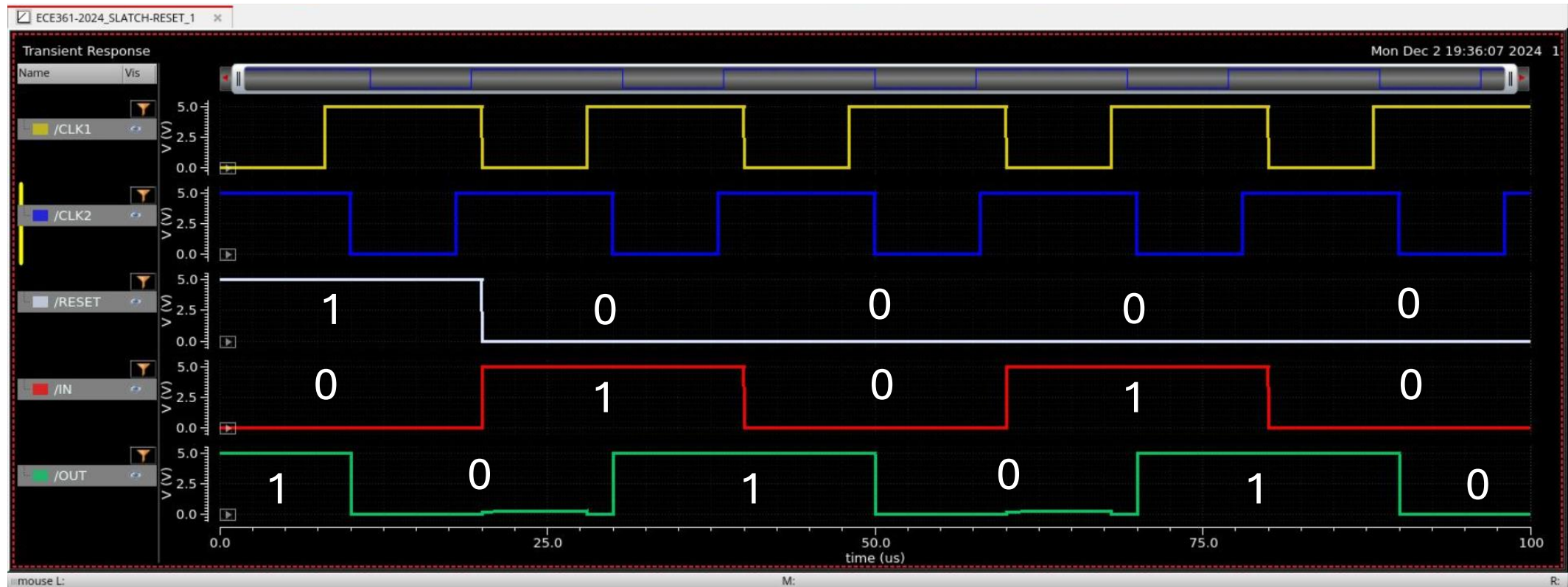
Adder Cell: Truth Table

A	B	Cin	SUM	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

1 Bit Shift Register - Static Latch Design

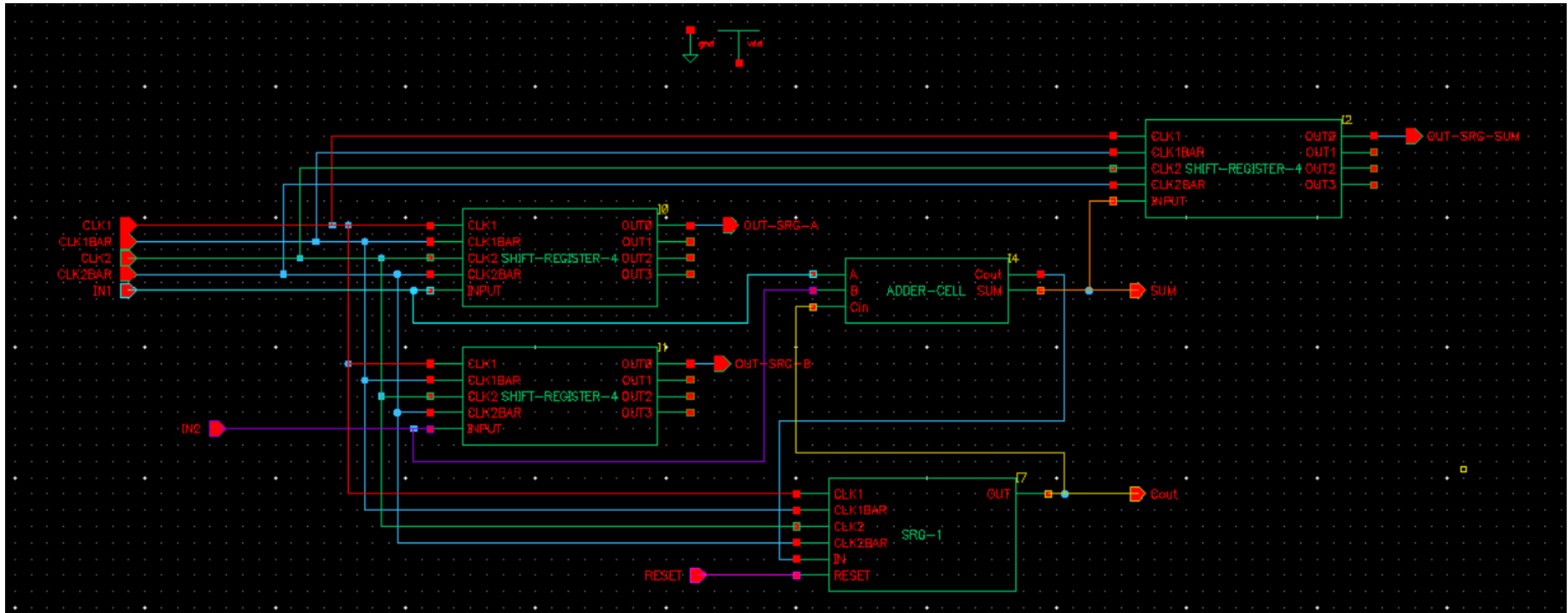
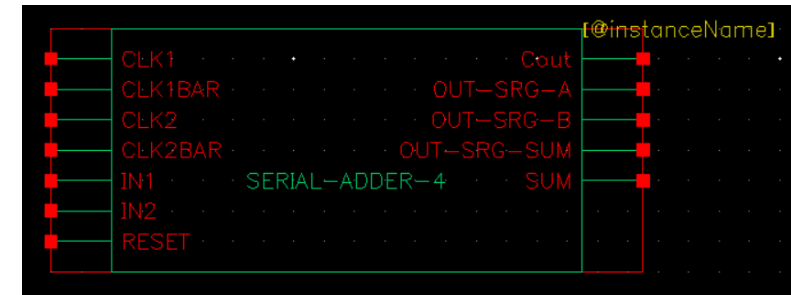


1 Bit Shift Register: Simulation

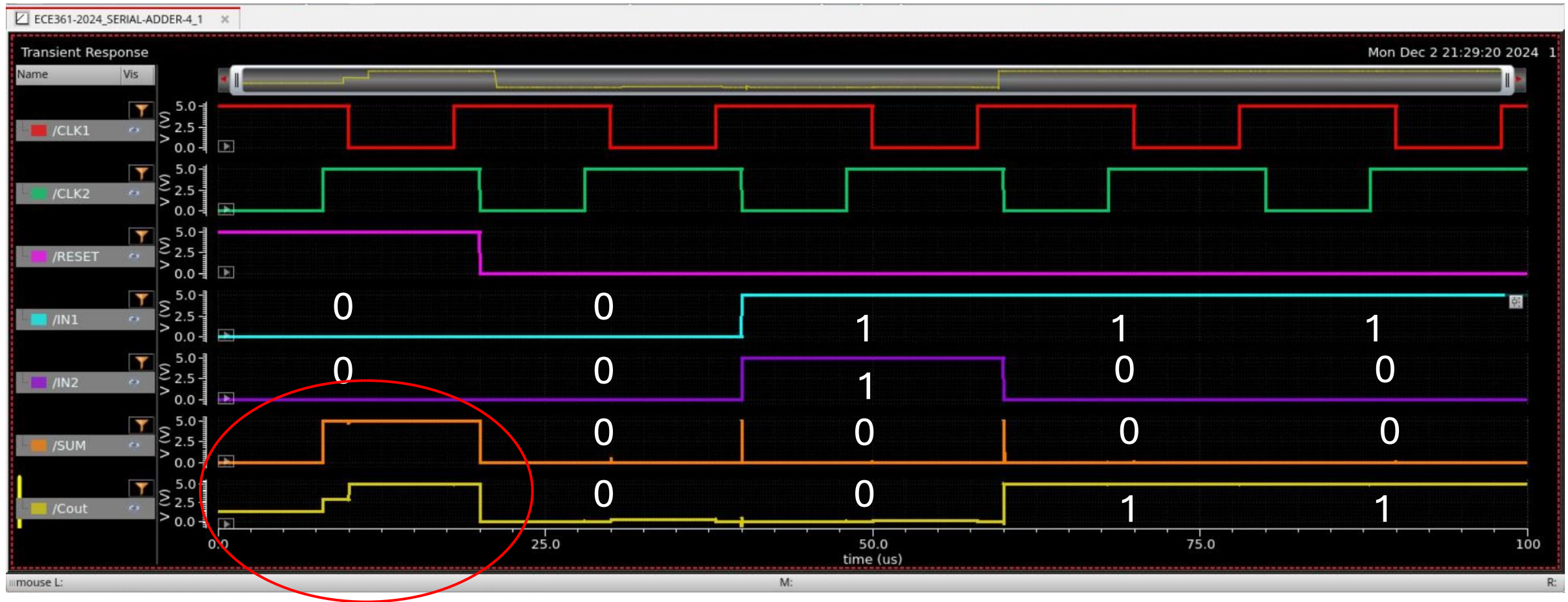


Output is the same as the Flip-Flop from before except there is a reset switch that outputs low whenever the reset switch is high. This is used to initialize the register with a low value.

Serial Adder

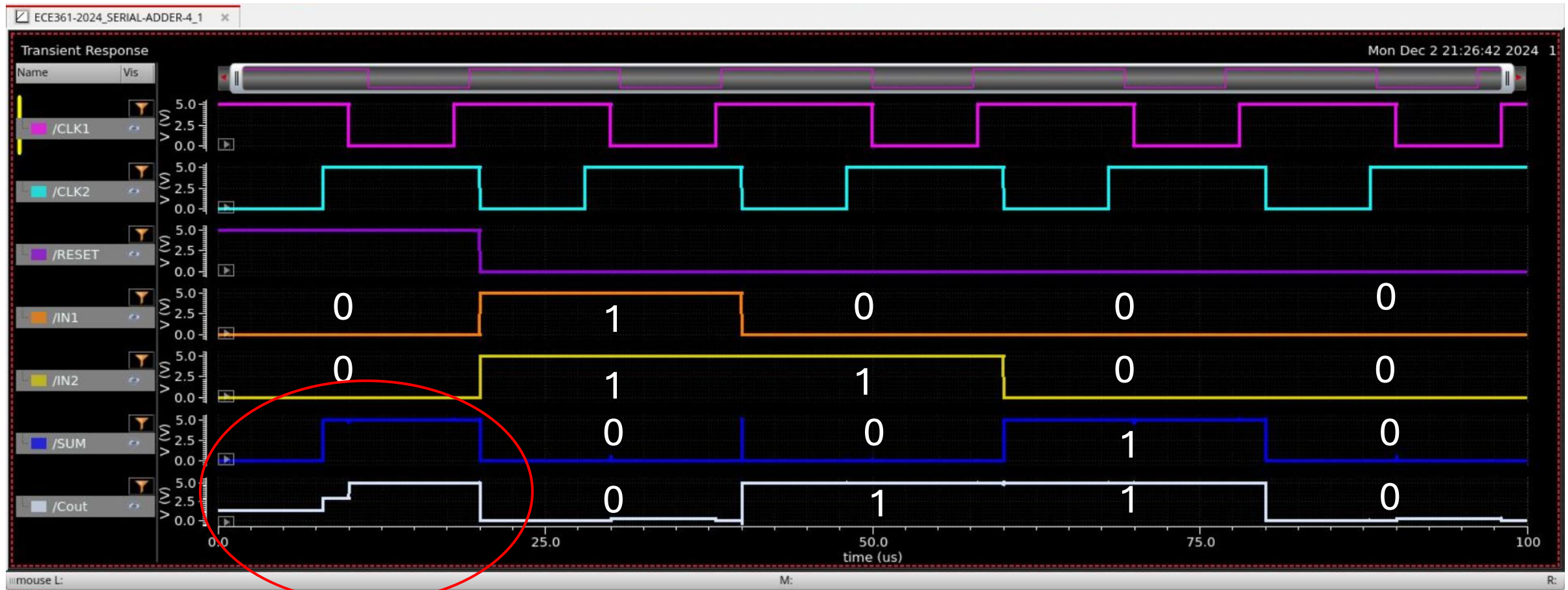


Serial Adder: Simulation 1



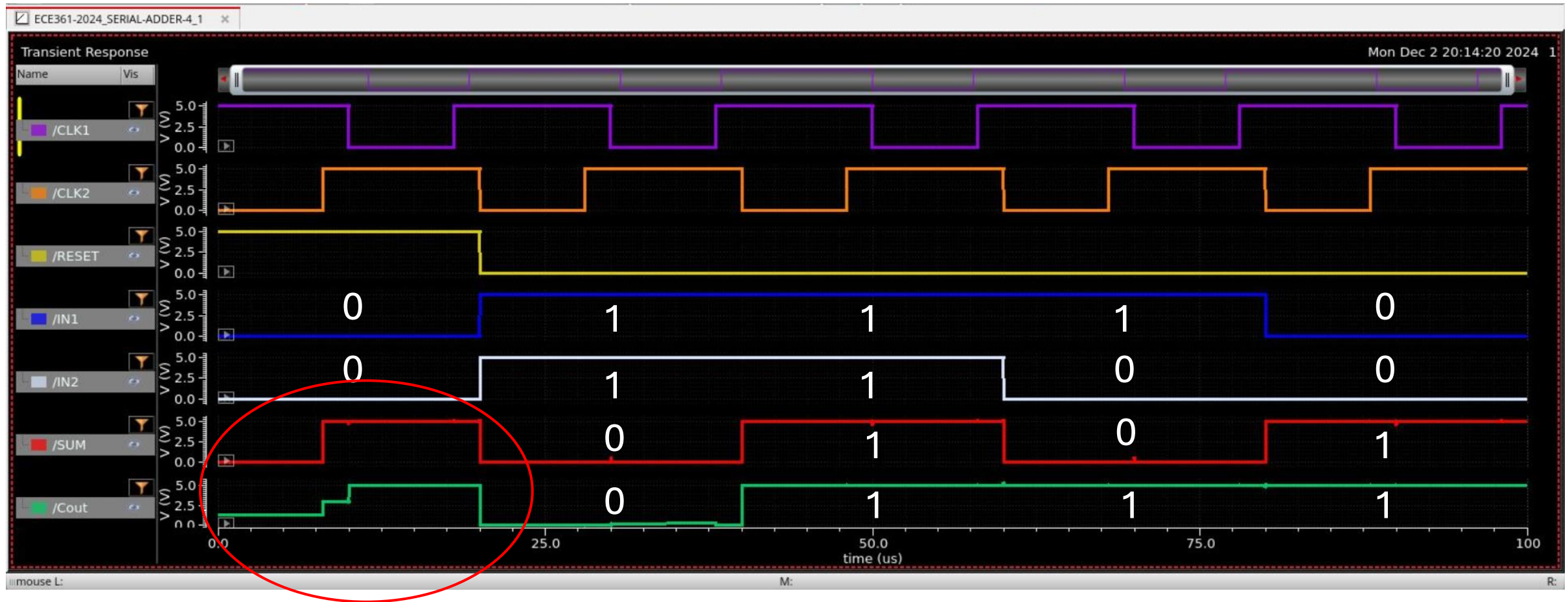
1110 + 0010 → Sum = 0000, Cout = 1 14 + 2 = 16 in base 10 (1110+0010 = 10000, is 5 bits long...)

Serial Adder: Simulation 2



0001 + 0011 → Sum = 0100, Cout = 0 1 + 3 = 4 in base 10

Serial Adder: Simulation 3



0111 + 0011 → Sum = 1010, Cout = 0 7 + 3 = 10 in base 10

Serial Adder: Discussion

- We can see that in each simulation the results match correctly with the binary addition.
- Also note that because the bits shift rightwards, the LSB is displayed first, and the MSB is displayed last in the simulation results.
- The red circles show that something weird is happening with initializing the 1 bit register that stores the carry out bit from the adder.
 - In the first clock cycle, while CLK2 is low (CLK1=High), the inner nodes are left floating, then CLK1 goes low and the first latch is set high. Once CLK2 goes low again (as the first bits are being added), the second latch is triggered and the register outputs low into the adder for the first carry in.
 - I am guessing that due to the initial floating nodes the voltage fluctuates that little bit